1. Problem Title & Source

Title: **Bank Loan Offer**

Source: **TCS NQT Previous Coding Questions**

2. Problem Statement (Clear Version)

A bank provides special **loan offers** to customers based on their financial history.

For each customer, the bank evaluates their **monthly income records**.

You are given:

- N → number of months
- arr[i] → income earned in the i-th month
- X → minimum income required to qualify for a loan

The bank will offer a loan **only if the customer has earned at least X income in any month**.

Your task is to determine **whether the customer qualifies for the loan offer**.

If the customer qualifies, print:

YES

Otherwise print:

NO

3. Examples

Example 1

Input

5

2000 3500 4000 1500 3000

3500

Output

YES

Explanation

One of the monthly incomes is:

3500

which meets the required minimum income.

Therefore the customer qualifies.

Example 2

Input

4

1000 2000 2500 3000

3500

Output

NO

Explanation

None of the months have income ≥ **3500**.

So the customer does not qualify for the loan.

4. Constraints

1 ≤ N ≤ 100

0 ≤ arr[i] ≤ 10^6

0 ≤ X ≤ 10^6

Observations

- We only need to check **whether any element meets the condition**
- No sorting or complex logic needed

5. Core Concept (Pattern)

Array Traversal + Condition Checking

Key idea:

Check if any arr[i] ≥ X

6. Thought Process

Step 1 — Read Input

We receive:

N

array of incomes

minimum required income X

Step 2 — Traverse Array

For each month:

if income ≥ X

  customer qualifies

Step 3 — Stop Early

Once we find a valid income:

print YES

Otherwise after checking all months:

print NO

7. Visualization

Example

arr = [2000, 3500, 4000, 1500, 3000]

X = 3500

Check incomes:

Income	Condition
2000	No
3500	Yes

Result

YES

8. Pseudocode

```
read N
read array arr
read X

for each income in arr

   if income ≥ X
      print YES
      stop

print NO
```

9. Code Implementation

Python

```
n = int(input())
arr = list(map(int,input().split()))
x = int(input())

for income in arr:
   if income >= x:
      print("YES")
      break
else:
   print("NO")
```

Java

```
import java.util.*;

class Main {

    public static void main(String[] args){

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int[] arr = new int[n];

        for(int i=0;i<n;i++)
            arr[i] = sc.nextInt();

        int x = sc.nextInt();

        for(int income : arr){

            if(income >= x){
                System.out.println("YES");
                return;
            }
        }

        System.out.println("NO");
    }
}
```

10. Time & Space Complexity

Time Complexity

O(N)

Worst case we check every element.

Space Complexity

O(1)

Only constant variables used.

11. Common Mistakes

1. Sorting the array unnecessarily
2. Checking all values even after finding valid income
3. Incorrect comparison (> instead of ≥)
4. Missing early termination

12. Detailed Dry Run

Input

N = 4

arr = [1000,2000,2500,3000]

X = 3500

Check

Income	≥ 3500
1000	No
2000	No
2500	No
3000	No

Result

NO

13. Real-Life Applications

Threshold checking appears in:

- Loan eligibility systems
- Credit score analysis
- Fraud detection
- Performance qualification checks

14. Common Traps

1. Using nested loops
2. Forgetting break condition
3. Reading input incorrectly
4. Comparing wrong variable

15. Related Problems

Threshold-based problems:

LC 1346 Check If N and Its Double Exist

LC 1431 Kids With the Greatest Number of Candies

LC 217 Contains Duplicate

16. Alternate Approaches

Approach	Complexity	Notes
Linear scan	O(N)	optimal
Sorting	O(N log N)	unnecessary

17. Why This Solution Works

Since the condition only requires **one valid month**, a single traversal of the array is enough.

The moment we encounter an income meeting the requirement, we can immediately conclude the result.

18. Variations / Follow-Up Questions

Possible interview variations:

1. Count months with income ≥ X
2. Find maximum income
3. Check if **average income ≥ X**
4. Loan eligibility with **multiple conditions**